

Riya Goyal

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EDUCATION

New Jersey Institute of Technology | Newark, NJ

Sep 2025 - May 2030

PhD in Mathematical Sciences (Track: Applied Mathematics), Official Joint degree from NJIT/Rutgers-Newark

Advisor: Dr. David Shirokoff

Relevant Coursework: Advanced Applied Math Modeling, Methods (ODE, PDE), Linear Algebra, Teaching in Math, Numerical Analysis, Dynamical Systems, Responsible Conduct of Research

Stockton University | Galloway, NJ

GPA: 3.99 (Dean's List: Fall 2021 to Spring 2025)

Sep 2021 - May 2025

Bachelor of Science in Mathematics with program distinction (Research with Faculty Member) and Honors distinction

Bachelor of Science in Computer Science with program distinction and Honors distinction

Minors in Business Administration, Computer Information Systems, and Digital Literacy/Multimedia Design

Study Abroad: F-1 student since 2021, 10-day Language & Cultural Program at National Taipei University of Education, Taiwan.

SKILLS

Language Skills: English (Native or Bilingual), Hindi (Native or Bilingual), Punjabi (Limited).

Technical Skills: *Programming:* Java, Python, JavaScript, React, HTML, CSS, PHP, R | *Database:* Oracle SQL, MySQL | *Other Tools:* LaTeX, Maple, Mathematica, MATLAB, Git, VMix, OBS, Visible Analyst, MS Office, Canva, WordPress.

PUBLICATIONS

- Raisa, T., Wei, D., **Goyal, R.**, & Ferri, C. (2026). Evaluating Agentic AI vs. Human Performance in ADRD Screening Literature Reviews. Accepted for GSA Annual Scientific Meeting, National Harbor, MD (Nov 4-7, 2026).
- Goyal, R.**, Henry, Y., Raisa, T., Fleck, J., & Wei, D. (2025). A Systematic Approach to Building High-Impact Aging Datasets for Researchers. *Innovation in Aging*, 9(Suppl 2), igaf122.3085. <https://doi.org/10.1093/geroni/igaf122.3085>.
- Wei, D., **Goyal, R.**, Raisa, T., Elmasri, J., & Fleck, J. (2025). GenAI Outperforms Human Reviewers in Classifying Alzheimer's Disease and Related Dementia Research. *Innovation in Aging*, 9(Suppl 2), igaf122.2250. <https://doi.org/10.1093/geroni/igaf122.2250>.
- Wei, D., **Goyal, R.**, Raisa, T., Elmasri, J., & Fleck, J. (2025, November 15-19). GenAI vs. Human Reviewers in Classifying Alzheimer's Disease and Related Dementia (ADRD) Literature: A Comparative Study. Presented at American Medical Informatics Association (AMIA) Annual Symposium, Atlanta, GA.
- Forrest B. & **Goyal R.** (2025). Lights Out on Nearly Complete Graphs. <https://doi.org/10.48550/arXiv.2508.09341>.
- Wei, D., Elmasri, J., Hernandez, B., **Goyal, R.**, & Burdick, D. (2024). Exploring the landscape of Generative AI and Large Language models in Alzheimer and Dementia's research. *Innovation in Aging*, 8(Suppl 1), 1112–1113. <https://doi.org/10.1093/geroni/igae098.3571>. Presented at the GSA Annual Scientific Meeting, Seattle, WA (Nov 13-16, 2024).
- Wei, D., Elmasri, J., **Goyal, R.**, Hernandez, B., & Burdick, D. (2024, November 9-13). Google Gemini-assisted scoping review of generative AI and large language models in ADRD research. Presented at American Medical Informatics Association (AMIA) Annual Symposium, San Francisco, CA.
- Wei, D. H., **Goyal, R.**, Prakash, S., & Zhang, R. (2023). Does COVID-19 affect mental health and substance use in young adults? *Journal of education and health promotion*, 12, 108. https://doi.org/10.4103/jehp.jehp_1167_22.

ACADEMIC AND TECHNICAL SERVICE

Abstract Reviewer, Gerontological Society of America (GSA)

- Served as a peer reviewer for the 2026 Annual Scientific Meeting.
- Evaluated 5 abstracts under the theme, "Reinforcing Resilience in Aging Science, Research, and Education."

Alumni Selection Committee, NJ Governor's STEM Scholars

- Evaluated 81 applications to identify highly qualified candidates for New Jersey's premier STEM leadership program 2026.
- Assessed applicants' academic trajectories, extracurricular involvement, and demonstrated commitment to STEM disciplines

Silver Judge, Technovation Girls (2026)

- Evaluated 8 technology-based projects for the 2026 program season.
- Provided personalized and actionable feedback to enrich the learning journeys of young tech innovators.

Event Judge, Newark College of Engineering First Year Engineering Design Showcase, NJIT (2026)

- Evaluated foundational engineering projects and formal presentations for first-year undergraduate students.
- Provided critical assessments to recognize and encourage student creativity, innovation, and technical design.

RESEARCH EXPERIENCE

Researcher in Health Analytics Research Team

Feb 2022 - Present

- Key member of the research team focused on advancing medical and health informatics by developing data-driven methodologies and AI-based tools including agentic AI frameworks to bridge the communication gap between physicians and patients. Contributing to research at the intersection of medical and health informatics through data-driven methodologies and AI-based tools.

- Key highlights include co-authored a publication in the Journal of Education and Health Promotion (2023), presented research at the GSA Annual Scientific Meeting and AMIA Annual Symposium (Nov 2024, Nov 2025), developed the website for the Stockton Center on Successful Aging (SCOSA) research initiative, leading the creation of a searchable, curated resource of medical and aging 100+ health datasets, currently finalizing a manuscript for a systematic review publication.

Undergraduate Researcher at Stockton University

Feb 2022 - August 2025

- Researched graph theory with Professor Bradley Forrest, analyzing *Lights Out* on random graphs. Developed algorithms using linear algebra and combinatorics, co-authored and submitted findings to *Involve* journal.

Council for Advancement and Support of Education (CASE) Intern

June 2024 - Aug 2024

- Conducted research at Scripps College on leveraging Artificial Intelligence tools for advancement, focusing on philanthropy.
- Performed peer research on 30+ institutions to analyze strategies and objectives for philanthropy efforts.

PRESENTATIONS

- **A Systematic Approach to Building High-Impact Aging Datasets for Researchers**, GSA Annual Scientific Meeting, Boston, MA, November 12-15, 2025.
- **GenAI Outperforms Human Reviewers in Classifying Alzheimer's Disease and Related Dementia Research**, Innovation in Aging GSA Annual Scientific Meeting, Boston, MA, November 12-15, 2025.
- **Exploring AI's Role in Mental Health Screening, Diagnosis, and Intervention for Older Adults: A Scoping Review**, Celebration of Scholarship and Teaching, Stockton University, April 7, 2025.
- **Curating High-Impact Secondary Datasets for Aging Research: A Systematic Review**, Stockton Center of Successful Aging Celebration of Aging, April 4, 2025.
- **Lights Out on a Nearly Random Complete Graph**, Mathematics Seminar, Stockton University, April 15, 2025; Undergraduate Mathematics Conference, Moravian University, February 15, 2025, Joint Math Meeting, January 10, 2025; Mathematics Seminar, Stockton University, November 25, 2024; Undergraduate Mathematics Conference, Moravian University, February 17, 2024; Mathematics Seminar, Stockton University, April 1, 2024.
- **Artificial Intelligence in Mental Health for the Aging Population** (Peer Presenter Badge), Celebration of Scholarship and Teaching, Stockton University, April 25, 2024.
- **Does COVID-19 affect mental health and substance use in young adults?** Celebration of Scholarship and Teaching, Stockton University, April 24, 2023.
- **Euler's Formula in Polyhedrons**, Mathematics Seminar, Stockton University, April 10, 2023.

TEACHING EXPERIENCE

New Jersey Institute of Technology | Newark, NJ

Teaching Assistant

Aug 2025 - Present

- Lead weekly recitations for two Calculus I sections (74 students total), reinforcing lecture material and facilitating problem-solving practice.
- Conduct quizzes and interactive exercises, providing individualized support to improve student comprehension and engagement.
- Facilitated the administration of undergraduate mathematical examinations for large-enrollment courses, ensuring a secure and standardized testing environment.

MATLAB Tutor and Grader

Jan 2026 - May 2026

- Acted as the designated point of contact for advanced MATLAB inquiries, debugging code, and explaining complex algorithms to students.
- Proactively roved the tutoring center to identify students in need of assistance, fostering an accessible and supportive learning environment.
- Serving as the grader for Linear Algebra course, helping with grading MATLAB assignments for undergraduate students.

Stockton University | Galloway, NJ

Educational Opportunity Fund (EOF) Math Tutor

June 2023 - Aug 2023, June 2024 - Aug 2024, June 2025 - Aug 2025

- Provided individualized math tutoring for 150+ students, enhancing students' understanding and performance.
- Organized workshops and study groups to support skill development beyond tutoring.

Mathematics and Computer Science Tutor

Sep 2022 - May 2025

- Developed tutoring strategies to enhance students' understanding of mathematical and computer science concepts.
- Provided one-on-one tutoring to 100+ students in mathematics and computer science, providing guidance and support.

Accessibility/Technology Tutor for Accommodated Staff

Sep 2023 - April 2024

- Provided individualized MS Office and Adobe software tutoring to a staff member, focusing on Microsoft Word, Excel, Outlook, and Adobe Acrobat, improving the staff member's productivity by 40%.
- Conducted weekly one-hour sessions, delivering instruction on various aspects of software use, including navigation, formatting, data management, and document editing.

Teaching Assistant / Grader for Discrete Mathematics

Sep 2023 - Dec 2023

- Assisted in teaching Discrete Mathematics by grading assignments, providing feedback, and supporting 30+ students.
- Collaborated with faculty to ensure fair grading while identifying common areas where students needed additional support.

ADDITIONAL EXPERIENCE

Stockton University | Galloway, NJ

Information Security Analyst

May 2025 - Aug 2025

- Assisted in security investigations by monitoring and analyzing 50,000+ logs across SIEM platforms, including Splunk, while gaining expertise in enterprise systems like Banner, Odyssey, and Argos to assess data flow, and security risks.

- Supported identity and access management initiatives for 3000+ users leveraging Duo Security for MFA, while managing security documentation and workflows using Jira and Confluence, improving response efficiency by 25%.

Resident Assistant

Aug 2022 - Aug 2025

- Assisted in the development and implementation of campus-wide programs to promote student engagement and success reaching 400+ residents.
- Selected as a **Lead Resident Assistant in Sep 2024** and helped Graduate Coordinator oversee 16 resident assistants.
- Participated in regular training sessions to stay updated on campus policies and procedures.

Switchboard Operator for Office of Information Technology

April 2024 - May 2025

- Provided efficient and courteous telephone assistance to university staff, faculty, and students.
- Directed 1000+ calls to appropriate departments, enhancing communication flow within the university.
- Developed strong multitasking abilities and gained valuable experience in a fast-paced office environment.

Admissions Ambassador

Oct 2021 - May 2025

- Provided personalized 150+ campus tours for prospective students, tailoring the tour to highlight specific programs.
- Assisted with daily office tasks in the Admissions Office and utilized technical skills to support the office.
- Selected as a **Lead Admissions Ambassador in May 2023** and trained and oversaw 30+ peer ambassadors.
- Supported Peer Mentoring for prospective students as one of two International Student Ambassadors, contributed to 20% increase in international student enrollment.

Esports Intern / Broadcast Engineer

Sep 2023 - Dec 2023

- Assisted in executing esports events to promote the university's esports program and attract new students.
- Provided tech support and streamed games online and assisted in 5+ large event gatherings for the program.

Interviewer at Polling Institute

Sep 2021 - Apr 2022

- Maintained a high level of professionalism and objectivity during 150+ interviews to ensure accurate and unbiased data collection and conducted in-depth research on polling trends to stay updated on the latest industry knowledge.

NJ Governor STEM Scholars Program

Sep 2024 - May 2025

Team Lead and Research Scholar

- Selected as Team Lead for the highly competitive Governor's STEM Scholars program (10% acceptance rate).
- Lead a team of 6 scholars on a STEM research project addressing community issues and presented findings to 50+ leaders.
- Engaged with New Jersey's STEM leaders and explored the state's STEM economy.

inMorphis Services Private Limited (Remote)

Associate Software Engineer

July 2021- May 2023

- Customized and configured ServiceNow applications to meet project requirements, conducted testing activities, including unit and integration testing, to ensure the quality and reliability of developed solutions, improving efficiency by 25%.
- Worked closely with cross-functional teams, including developers, system administrators, and stakeholders, to ensure project success, identified and resolved issues within the platform by providing timely resolutions.

PROFESSIONAL DEVELOPMENT

- OurFa2M2 Conference (*Feb 7-8, 2026*)
- GSA Annual Scientific Meeting (*Nov 12-15, 2025*)
- NJ Governors STEM Commencement Conference at Kean University (*May 3, 2025*)
- Undergraduate Mathematics Conference at Moravian University (*Feb 16, 2025, Feb 17, 2024; Feb 18, 2023*)
- NJ Governors STEM in Academia Conference Montclair State University (*January 25, 2025*)
- Joint Math Meeting (*Jan 8-11, 2025*)
- NJ Governors STEM in Government Conference at Rutgers University (*November 9, 2024*)
- HackHarvard (*Oct 11-13, 2024*)
- GROW Conference at Columbia University (*Sep 27-29, 2024*)
- CASE Conference (*June 23-25, 2024*)
- Inclusive Leadership Conference at Stockton University (*Apr 5, 2024*)
- NJ Governors Team Lead Meeting at Princeton University (*September 20, 2024*)
- Garden State Undergraduate Mathematics Conference at Kean University (*Apr 1, 2023*)

AWARDS

- NJIT Presidential Leadership Award (*2026*)
- Graduate Student Appreciation Award for my role as Department Representative (*2026*)
- Certificate of Service as Cybersecurity Practitioner (*2025*)
- International Computer Science Competition (Senior Category): International Rank 3 (*2025*)
- Kiwanis Children's Fund Scholarship (CKI) (*2025*)
- Alpha Lambda Delta Awards/Fellowships:
 - Graduate Fellowship (*2025*)
 - Jo Anne J Trow Undergraduate Scholarship Recipient (*2023*)
 - First Year Excellence Award (*2022*)
- Certificate of Achievement in MCM - Mathematical Modeling Contest (*2023, 2024, 2025*)
- Success Scholar Awards:
 - Outstanding Group Leader & Success Scholar (*2025*)
 - Outstanding Group Leader & Workshop Presenter Award (*2024*)
 - Dedicated Peer Mentor Award (*2023, 2022*)

- Stockton Development Awards:
 - L.E.A.D. Award (2025)
 - Student Leader of the Year Award (2025)
 - Spirit Award (2024)
 - Engagement Award (2023)
- Stockton Community Engagement Student Award (2025)
- Stockton Board of Trustees Fellowship Recipient (2024)
- Circle K Carter Family Scholarship Recipient (2024)
- First to Fly Award (2024)
- Top 100 Influential Indian Award by Hypedge Media (2023)
- Young Achievers Award by India Proud Book (2023)
- Young Motivator and Influencer Award by London Book of World Records (2023)
- National Education and Student Excellence Awards by Plugin PR and Marketing Group (2023)
- Asia Excellence Award by MTTV News Media Pvt. Ltd. (2023)
- Pi Mu Epsilon Superior Achievement in Math (2023)
- International Youth Math Challenge (IYMC):
 - Silver Honor (2024, 2023)
 - Gold Honor (2022)
 - US Topper (2021)
- Sipra Pal Award for Perseverance in Physics (2022)

LEADERSHIP POSITIONS

- Alternate Department Representative of Mathematical Sciences for Graduate Student Association (*Jan 2026-Present*)
- NJIT Graduate Women in Computing (GWICS): Treasurer (*May 2026-Present*), Co-Treasurer (*Nov 2025-May 2026*)
- NJIT SIAM: Vice President (*Sep 2025-Present*)
- Sanskar:
 - Secretary (*May 2026-Present*)
 - Associate (*Sep 2025-May 2026*)
- OurFa2M2 Conference: Co-Organizer (*June 2025-Present*)
- Stockton Young Alumni Club (YAC) Advisory Board (*May 2025-Present*)
- Circle K Garden State Young Professionals: Member (*May 2025-May 2026*)
- Student Senator:
 - Academic Affairs Committee (*May 2024-May 2025*)
 - Diversity and Inclusion Committee (*Sep 2023-May 2024*)
- Residential Scholar (*Sep 2023-May 2025*)
- Stockton International Student Organization (ISO): Founding President (*Oct 2022-May 2025*)
- Stockton Honors College:
 - President (*Jan 2023-May 2025*)
 - Peer Mentor (*Sep 2022-May 2025*)
 - Director of Fundraising and Finances (*Jan 2022-Dec 2022*)
- Student Success Scholars Program: Peer Mentor (*Sep 2022-May 2025*), Group Study Leader (*Sep 2023-May 2025*)
- School of Business: Student Advisory Board Member (*Oct 2021- May 2025*)
- Alpha Lambda Delta National Honors Society:
 - President (*June 2023- May 2025*)
 - Vice President (*Sep 2022-May 2023*)
- Stockton Computer Society: Vice President (*Sep 2022-May 2023*)
- Campus Hearing Board: Member (*Oct 2021-May 2022*)

ASSOCIATIONS & MEMBERSHIPS

- Gerontological Society of America (GSA)
- NJ Governor STEM Scholars Program
- Advancing Indigenous People in STEM (AISES)
- Aspire Leaders Program by Harvard University
- Circle K International: Society of Distinguished Collegians
- Foundations of Computational Mathematics (FoCM)
- Alpha Lambda Delta (ALD) - Honors Society
- Golden Key: National Honors Society
- Pi Mu Epsilon (PME): National Maths Honors Society
- Society of Actuaries (SOA)
- Society for Industrial and Applied Mathematicians (SIAM)
- Union Council of Academics (UCAN) at NJIT

CERTIFICATIONS

- First lessons with MakeCode and micro: bit (*Jan 2026*)
- OpenCV Bootcamp (*Dec 2025*)
- J.P. Morgan - Quantitative Research Job Simulation (*Dec 2025*)
- RCR Basic Course by CITI Program (*Sep 2025*)

- Accenture Nordics - Software Engineering Job Simulation (*Jan 2025*)
- ChatGPT for Developers by UpGrad (*Aug 2024*)
- Elements of AI by University of Helsinki (*Aug 2024*)
- Business Intelligence using Power BI by Skill Nation (*July 2024*)
- Ethics in the Age of Generative AI (*July 2024*)
- Career Essentials in Generative AI by Microsoft and LinkedIn (*July 2024*)
- Data Science Mathematics by Great Learning (*July 2024*)
- Certified Lean Six Sigma AI Yellow Belt (*July 2024*)
- HackerRank Certification: SQL (*Jan 2024*)
- Micro Certification: Welcome to ServiceNow (*March 2023*)
- ServiceNow Certified System Administrator (*Dec 2022*)

COMMUNITY ENGAGEMENT

- Circle K International: Service Organization (*Sep 2021-Present*)
- Encorps Math Tutor Volunteer (*May 2024-Present*)
- SchoolHouse Math Tutor Volunteer (*Sep 2024-Present*)
- NJIT's Center for Pre-College Programs: Mentor for eSTEM Challenge (*Jan 2026- March 2026*)
- Stockton University Honors College (*Sep 2021- May 2025*)
- FreeRice (*May 2022- May 2025*)
- Stockton Global Engagement Mentor (GEM) (*June 2024- May 2025*)
- National Integration Forum of Artists and Activities (*2019-2021*)
- Akhil Bhartiya Aggarwal Samelan (Women Wing) (*2017-2021*)